SIMATS ENGINEERING
R PROGRAMMING – ASSESSMENT TOOL 1

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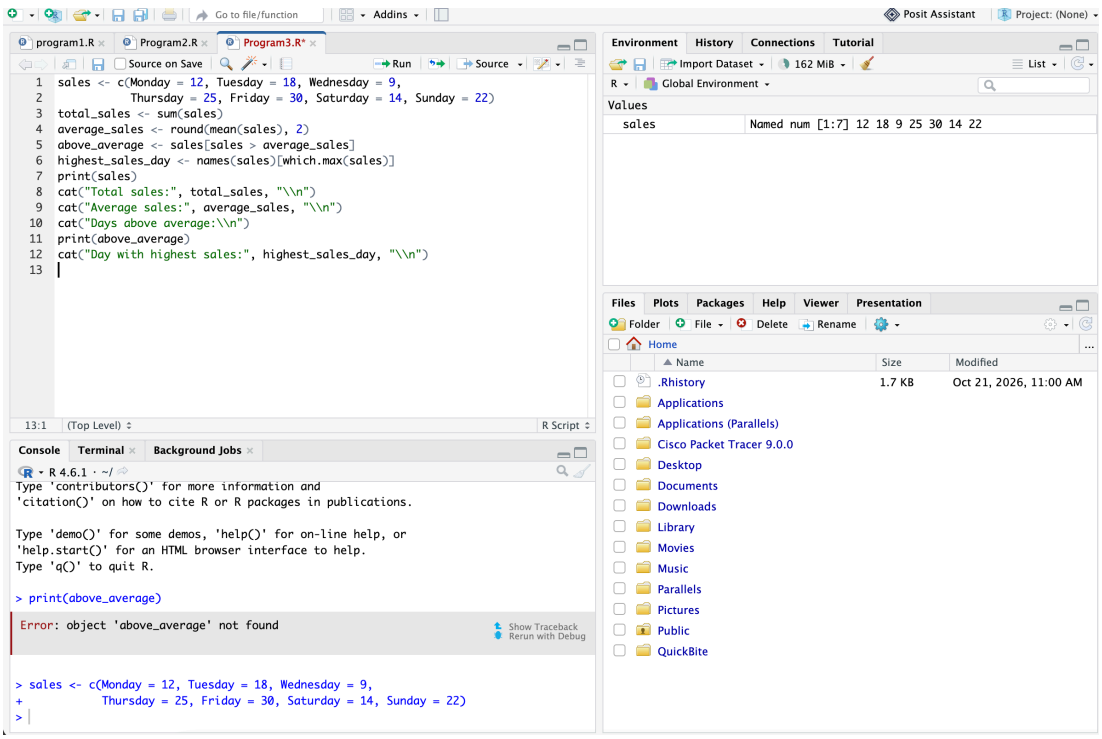
Reg.No:192525013

Question 1: Vectors – Weekly Sales Analysis

Aim: To create a named vector of weekly sales and calculate summary information using vector operations and logical indexing.

R Program

```
sales <- c(Monday = 12, Tuesday = 18, Wednesday = 9,
           Thursday = 25, Friday = 30, Saturday = 14, Sunday = 22)
total_sales <- sum(sales)
average_sales <- round(mean(sales), 2)
above_average <- sales[sales > average_sales]
highest_sales_day <- names(sales)[which.max(sales)]
print(sales)
cat("Total sales:", total_sales, "\\n")
cat("Average sales:", average_sales, "\\n")
cat("Days above average:\\n")
print(above_average)
cat("Day with highest sales:", highest_sales_day, "\\n")
```



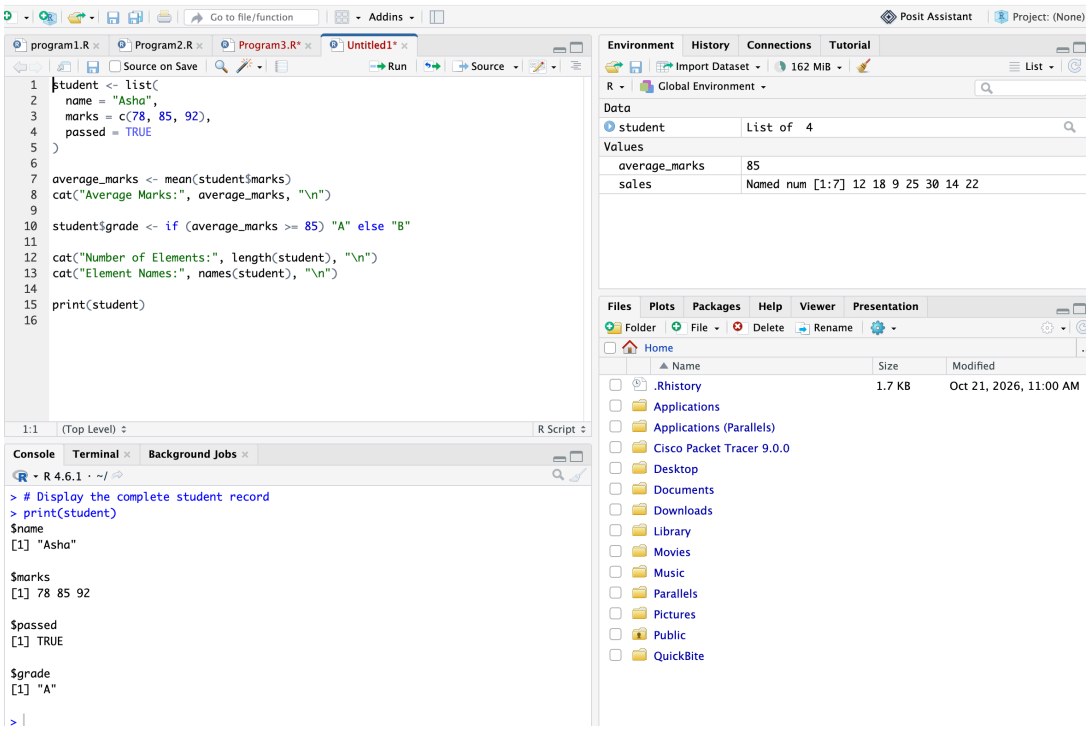
Question 2: Lists – Student Record

Aim: To store a student's name, test marks, and pass status in a list, then add a grade without rebuilding the list.

```
student <- list(
  name = "Asha",
  marks = c(78, 85, 92),
  passed = TRUE
)

average_marks <- mean(student$marks)
student$grade <- if (average_marks >= 85) "A" else "B"

cat("Average marks:", average_marks, "\\n")
cat("Number of elements:", length(student), "\\n")
cat("Element names:", names(student), "\\n")
print(student)
```



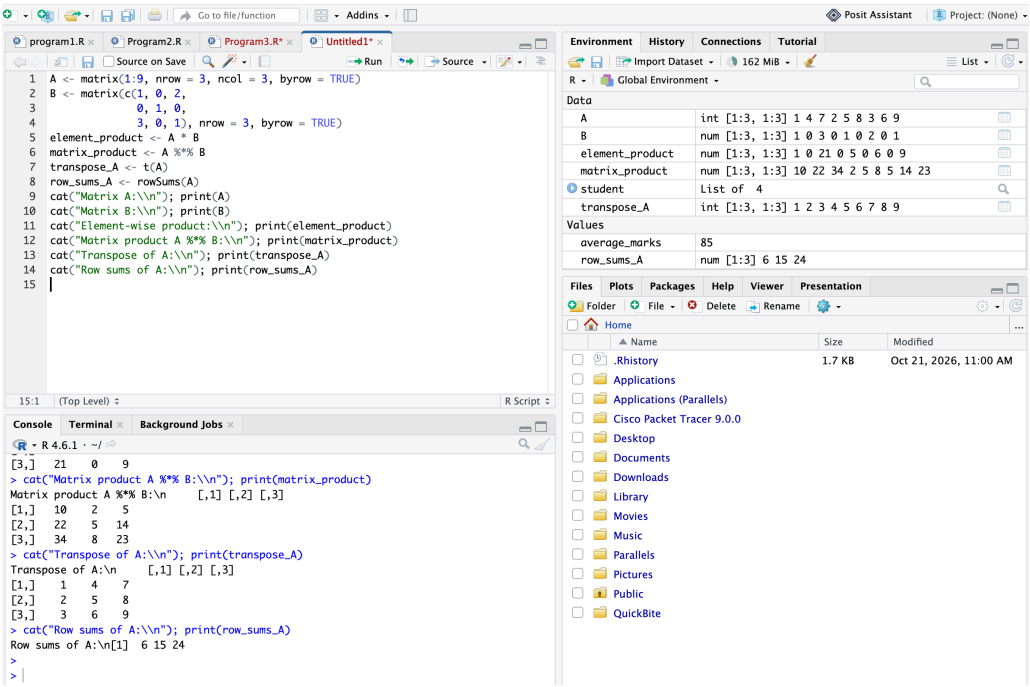
Question 3: Matrices – Operations on Two Matrices

Aim: To create two 3 × 3 matrices and perform element-wise multiplication, matrix multiplication, transposition, and row-sum calculations.

```
A <- matrix(1:9, nrow = 3, ncol = 3, byrow = TRUE)
B <- matrix(c(1, 0, 2,
              0, 1, 0,
              3, 0, 1), nrow = 3, byrow = TRUE)
```

```
element_product <- A * B
matrix_product <- A %*% B
transpose_A <- t(A)
row_sums_A <- rowSums(A)

cat("Matrix A:\\n"); print(A)
cat("Matrix B:\\n"); print(B)
cat("Element-wise product:\\n"); print(element_product)
cat("Matrix product A %*% B:\\n"); print(matrix_product)
cat("Transpose of A:\\n"); print(transpose_A)
cat("Row sums of A:\\n"); print(row_sums_A)
```



Question 4: Arrays – Three-Dimensional Data

Aim: To create a 3-dimensional array for warehouse item counts, extract specified values, and calculate totals for each layer using apply().

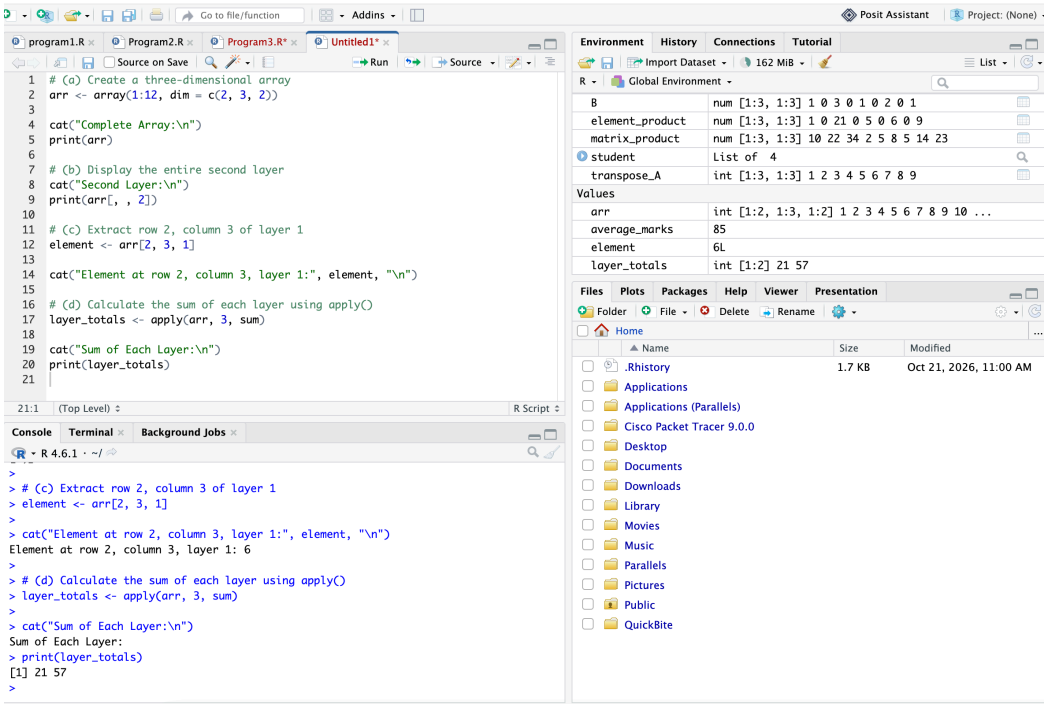
```
arr <- array(1:12, dim = c(2, 3, 2))

cat("Entire array:\\n")
print(arr)

cat("Second layer:\\n")
print(arr[, , 2])

cat("Element at row 2, column 3, layer 1:",
    arr[2, 3, 1], "\\n")

layer_totals <- apply(arr, 3, sum)
cat("Layer totals:\\n")
print(layer_totals)
```



Question 5: Class Types and Coercion – Price List

Aim: To convert character prices into numeric values, calculate the total and discounted bill, and demonstrate automatic type coercion in R.

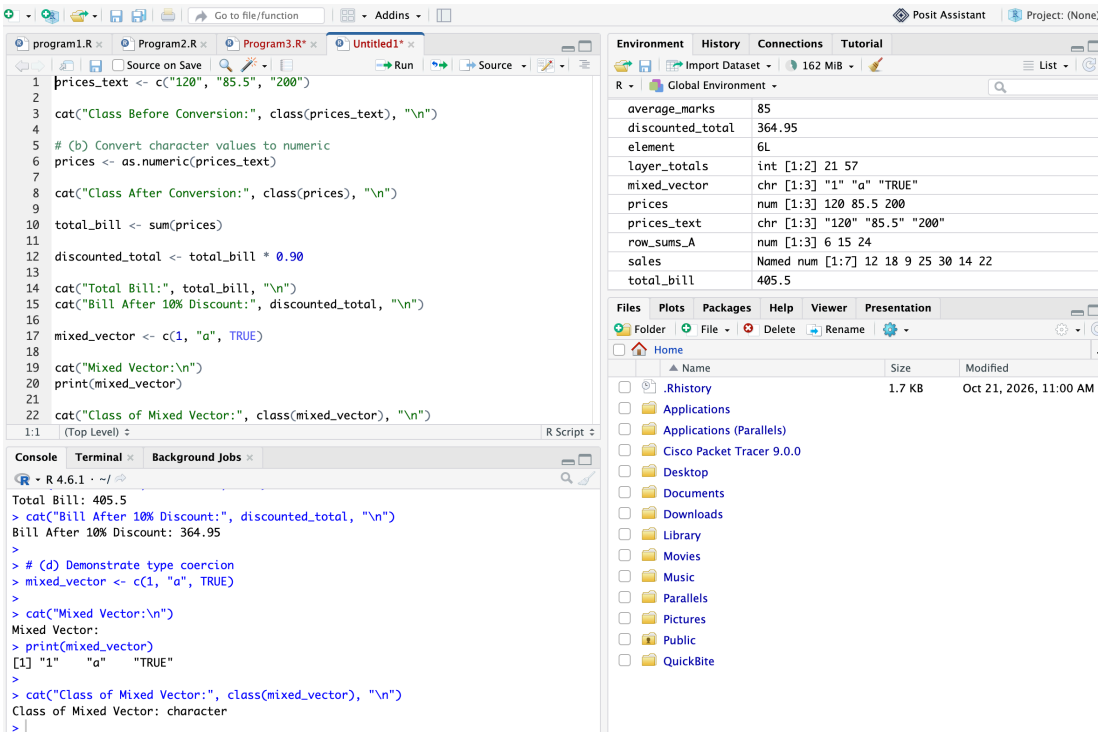
```
prices_text <- c("120", "85.5", "200")
cat("Class before conversion:", class(prices_text), "\\n")

prices <- as.numeric(prices_text)
cat("Class after conversion:", class(prices), "\\n")

total_bill <- sum(prices)
discounted_total <- total_bill * 0.90

cat("Total bill:", total_bill, "\\n")
cat("Bill after 10% discount:", discounted_total, "\\n")

mixed_vector <- c(1, "a", TRUE)
print(mixed_vector)
cat("Class of mixed vector:", class(mixed_vector), "\\n")
```


program1.R
Program2.R
Program3.R*
Untitled1*
Source on Save
Run
Source
prices_text <- c("120", "85.5", "200")
cat("Class Before Conversion:", class(prices_text), "\n")
(b) Convert character values to numeric
prices <- as.numeric(prices_text)
cat("Class After Conversion:", class(prices), "\n")
total_bill <- sum(prices)
discounted_total <- total_bill * 0.90
cat("Total Bill:", total_bill, "\n")
cat("Bill After 10% Discount:", discounted_total, "\n")
mixed_vector <- c(1, "a", TRUE)
cat("Mixed Vector:\n")
print(mixed_vector)
cat("Class of Mixed Vector:", class(mixed_vector), "\n")
Console
Terminal
Background Jobs
R 4.6.1 · ~/
Total Bill: 405.5
> cat("Bill After 10% Discount:", discounted_total, "\n")
Bill After 10% Discount: 364.95
> # (d) Demonstrate type coercion
> mixed_vector <- c(1, "a", TRUE)
> cat("Mixed Vector:\n")
Mixed Vector:
> print(mixed_vector)
[1] "1" "a" "TRUE"
> cat("Class of Mixed Vector:", class(mixed_vector), "\n")
Class of Mixed Vector: character
Environment
History
Connections
Tutorial
Import Dataset
162 MiB
List
Global Environment
average_marks 85
discounted_total 364.95
element 6L
layer_totals int [1:2] 21 57
mixed_vector chr [1:3] "1" "a" "TRUE"
prices num [1:3] 120 85.5 200
prices_text chr [1:3] "120" "85.5" "200"
row_sums_A num [1:3] 6 15 24
sales Named num [1:7] 12 18 9 25 30 14 22
total_bill 405.5
Files
Plots
Packages
Help
Viewer
Presentation
Folder
File
Delete
Rename
Home
Name
Size
Modified
.Rhistory 1.7 KB Oct 21, 2026, 11:00 AM
Applications
Applications (Parallels)
Cisco Packet Tracer 9.0.0
Desktop
Documents
Downloads
Library
Movies
Music
Parallels
Pictures
Public
QuickBite